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Characterisation, Field Modelling, and Radiation Testing of a Commercial Tunnelling Magnetoresistance (TMR) Sensor for CubeSat Attitude Determination and Control

Final Year Technical Report

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[TODO: insert submission date]

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Nitish, Aaron, Arnesh, Alex, jensen, Uzayr

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Abstract

Tunnelling magnetoresistance (TMR) sensors offer chip-scale, high-sensitivity magnetic field detection that is attractive for CubeSat Attitude Determination and Control System (ADCS) and Electrical Power System (EPS) applications, yet the technology remains at a low Technology Readiness Level (TRL 3) with limited published space-qualification data. This report documents the design, construction, and testing of two custom PCBs built around a commercial-off-the-shelf (COTS) CT100 TMR sensor: a primary board exposing the raw sensor bridge, and an integrated telemetry board adding STM32-based automated logging and an independent current-sense field reference. The CT100's Wheatstone-bridge topology and its saturation ceiling of $V_{CC}/3$ are derived directly from the device's specified per-element resistance range and confirmed experimentally. Sensitivity was quantified via a current-sweep method, cross-validated across three independent instruments, giving $3.7\,\mathrm{mV\,V^{-1}\,mT^{-1}}$ against the CT100's $4.5\,\mathrm{mV\,V^{-1}\,mT^{-1}}$ datasheet figure. The sensor was subjected to Total Ionising Dose (TID) X-ray radiation testing (320 kV, tungsten target, 3 mm beryllium filter, 10 mA); the same sensitivity method applied post-irradiation showed no statistically significant change (93.6–99.8% of the unirradiated value), and an initial ferrite-shielded enclosure showed a qualitative improvement toward simulated field predictions. **[TODO: Expand this abstract once shielded-sweep and radiation-plus-shielding quantitative results are finalised – see notes throughout this draft.]**

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Glossary of Terms

ADCS	Attitude Determination and Control System
ADC	Analogue-to-Digital Converter
AMR	Anisotropic Magnetoresistance
COTS	Commercial-Off-The-Shelf
EPS	Electrical Power System
FEM	Finite Element Method
FGM	Fluxgate Magnetometer
GMR	Giant Magnetoresistance
HAL	Hardware Abstraction Layer (STM32 firmware library)
I^2C	Inter-Integrated Circuit (serial communication bus)
INA	Current-sense amplifier (current monitor IC)
krad(Si), Mrad(Si)	Kilorad / megarad of absorbed dose in silicon – units of Total Ionising Dose
LDO	Low-Dropout (linear voltage regulator)
MCU	Microcontroller Unit
MgO	Magnesium Oxide (crystalline tunnel-barrier material)
MI	Magneto-Inductive (sensor technology)
MTJ	Magnetic Tunnel Junction
OBC	On-Board Computer
PCB	Printed Circuit Board
SWaP-C	Size, Weight, Power, and Cost
SWV	Serial Wire Viewer (ARM Cortex debug trace interface)
TID	Total Ionising Dose
TMR	Tunnelling Magnetoresistance
TRL	Technology Readiness Level
Rad-Hard	Radiation-Hardened

1 Introduction

CubeSats are nanosatellites (1–10 kg) built from standardised 10 cm cubic units, offering low-cost, rapid access to space relative to traditional monolithic satellite missions. This miniaturisation is achieved at the expense of severe Size, Weight, Power, and Cost (SWaP-C) constraints, which particularly affect the Attitude Determination and Control System (ADCS): the subsystem responsible for orienting a CubeSat for communications, power generation, and payload pointing. Onboard magnetometers are the primary means by which a CubeSat senses its attitude relative to the Earth's magnetic field, so the choice of magnetometer technology is central to ADCS performance within a tightly constrained mass, volume, and power budget.

Heritage fluxgate magnetometers offer excellent sensitivity but are too large and power-hungry for CubeSat platforms, motivating a shift toward chip-scale magnetoresistive sensors. Among these, Tunnelling Magnetoresistance (TMR) sensors offer the highest sensitivity of the commercially available options, in a footprint of only a few millimetres square. However, TMR remains comparatively immature for space use – typically cited at Technology Readiness Level (TRL) 3 – with comparatively little published characterisation data under space-relevant stresses such as strong transient fields from onboard actuators (e.g. magnetorquers) and ionising radiation.

This report documents the design, construction, and testing of a project evaluating a commercial-off-the-shelf (COTS) CT100 TMR sensor for CubeSat ADCS/EPS applications, carried out by the author. Two custom PCBs were developed: a primary board exposing the raw CT100 bridge output, and an integrated telemetry board combining the sensor with microcontroller-based logging and an independent current-based field reference. The work addresses four objectives:

(i) characterise the CT100 sensor itself – its operating principle, its Wheatstone-bridge topology, and its saturation and hysteresis behaviour;

(ii) design and build hardware capable of generating a known, controllable test field and logging the sensor's response to it automatically;

(iii) quantify the sensor's sensitivity, saturation-recovery time, and the accuracy of the field model used to drive it; and

(iv) assess the sensor's tolerance to Total Ionising Dose (TID) radiation representative of the CubeSat orbital environment, with and without magnetic shielding.

A parallel thermal-cycling test campaign investigating TMR temperature sensitivity and thermal-shock response was led by project partner Daniel Tieu and is reported separately; it is outside the scope of this report.

The remainder of this report follows the project's own design and testing sequence. Section 2 reviews the relevant literature. Section 3 sets out the resulting research question, aims, and objectives. Section 4 characterises the CT100 sensor itself. Section 5 introduces the instrumentation used throughout. Section 6 covers the standalone sensor board and sensor-only saturation testing. Section 7 covers the design of the integrated board and its field-excitation method, including validation of the field model against simulation. Section **??** cross-validates the instrumentation used throughout. Section 8 presents the non-irradiated sensitivity and saturation dataset. Section 9 covers radiation testing, and Section 10 covers magnetic shielding, including the combined radiation-plus-shielding result. Sections 12 and 13 discuss the findings and conclude.

2 Literature Review

2.1 Magnetometer Technologies for CubeSat ADCS

Fluxgate magnetometers (FGMs) have been used in space exploration for decades: an alternating current drives a coil-wrapped permeable core toward saturation, and an external field produces a measurable imbalance between drive and sense coil currents. FGMs offer excellent sensitivity (of order $10\,\mathrm{pT}/\sqrt{\mathrm{Hz}}$) but are far too large and power-hungry for CubeSat platforms, typically weighing over 1 kg and drawing more than 10 W [2]. Magneto-inductive (MI) sensors, which sense an external field through the change it induces in the resonant frequency of an LR oscillator, trade sensitivity for a much smaller footprint and lower power draw [2].

Hall-effect sensors are inexpensive and, because they contain no ferromagnetic material, immune to magnetic saturation. However, their comparatively low sensitivity confines them to detecting fields above roughly 1 mT, and while flux concentrators can improve their effective sensitivity, doing so reintroduces the mass and volume penalty that SWaP-C constraints are meant to avoid [3]. Magnetoresistive sensors – AMR, GMR, and TMR – instead occupy the sub-milliTesla sensitivity range needed for geomagnetic-field sensing while remaining chip-scale. AMR sensors (typically permalloy thin films) achieve high resolution and bandwidth but saturate above roughly 1 mT and often require periodic magnetic reset [3]. GMR sensors, based on a multilayer stack of alternating ferromagnetic and non-magnetic conductive layers, offer higher sensitivity again but are more prone to hysteresis-induced offset and can be permanently degraded by strong fields [3]. Chip-scale magnetoresistive sensing has already reached flight heritage: the AMR-based MAGIC instrument, for example, has demonstrated a 104 g, 0.5 W dual-sensor vector magnetometer in orbit on the RadCube CubeSat [4].

TMR sensors extend this trend further, as detailed for the specific device used in this project in Section 4.

2.2 Radiation Tolerance of TMR Sensors

Because TMR is comparatively new to the space sector, published radiation-tolerance data is limited, though what exists is promising. Alves' assessment of magnetoresistive sensors for space applications supports the conclusion that TMR devices are radiation-hard [5]. Separately, published testing of TMR magnetic sensors based on MTJ structures found their performance unchanged under bias voltages of $+5$, 0, and -5 V after a Total Ionising Dose (TID) of 5 Mrad(Si) using γ-ray and X-ray sources [6] – two orders of magnitude beyond the 100 krad(Si) dose typically used to qualify a device as radiation-hardened. This is physically consistent with the crystalline MgO barrier at the heart of the MTJ: ionising radiation predominantly damages devices through electron-hole pairs that become trapped at defect sites, permanently shifting threshold voltages – a mechanism that primarily affects the *amorphous* SiO_2 gate oxide used in conventional CMOS devices [7], and for which MgO's ordered crystal lattice offers far fewer trap sites. This literature evidence, and the tolerance mechanism it points to, motivated the TID testing described in Section 9 of this report. **[TODO: The specific X-ray energy/exposure combination used in Section 9 has not been located elsewhere in the TMR-sensor radiation literature during this review; confirm this before claiming novelty in Section 9/12, and cite accordingly if a search turns up nothing.]**

2.3 Research Gap

Taken together, this literature establishes three things: TMR is the most sensitive chip-scale magnetoresistive technology available for CubeSat-class ADCS/EPS use (Section 2.1); its radiation tolerance, while under-studied compared to AMR/GMR, is theoretically well-motivated and supported by limited existing testing at the device level (Section 2.2); but it remains at a low technology readiness level, and published characterisation data addressing space-*operational* conditions – specifically, behaviour under the strong transient fields produced by an onboard magnetorquer, and behaviour after ionising-radiation exposure of a specific, low-cost COTS part – is largely absent for TMR relative to AMR/GMR. **[TODO: Strengthen this paragraph once your search for directly comparable TMR magnetorquer-saturation and TID studies (see the novelty flag in Section 2.2) is complete; if genuinely no directly comparable study exists, say so explicitly here, since an explicit identification of an evidence gap is itself part of what this rubric criterion is assessing.]** This gap, evidence at the level of a real, specific COTS device under space-relevant transient and radiation stress, rather than the general TMR technology class, motivates the research question and objectives set out in Section 3.

3 Research Question, Aims, and Objectives

Research question. Is a low-cost, commercial-off-the-shelf TMR sensor (the CT100) operationally viable for CubeSat ADCS/EPS use, specifically with respect to (a) its behaviour when driven into and recovered from magnetic saturation by a transient field comparable to an onboard magnetorquer pulse, and (b) any change in its sensing performance following Total Ionising Dose radiation exposure representative of the CubeSat orbital environment?

This question follows directly from the gap identified in Section 2.3: the literature supports TMR's theoretical suitability and likely radiation tolerance, but does not provide direct, device-level evidence of either property for a specific, inexpensive COTS part of the kind a resource-constrained CubeSat programme would actually select.

Aims. This project aims to generate that device-level evidence for the CT100, and, in doing so, to build and validate a low-cost, automatable test methodology that could be reused for other candidate COTS TMR devices without requiring specialised space-qualification equipment.

Objectives. The research question and aims above translate into the following concrete objectives, which structure the remainder of this report:

(i) characterise the CT100 sensor itself – its operating principle, its Wheatstone-bridge topology, and its saturation ceiling (Section 4);

(ii) design and build hardware capable of generating a known, controllable test field and logging the sensor's response to it automatically, without relying on specialised or costly space-test equipment (Sections 5–7);

(iii) quantify the sensor's sensitivity, saturation-recovery time, and the accuracy of the field model used to drive it, establishing a validated non-irradiated baseline cross-validated across instruments (Sections **??**–8);

(iv) evaluate whether low-cost magnetic shielding materially changes the accuracy of that baseline characterisation (Section 10); and

(v) assess the sensor's tolerance to Total Ionising Dose radiation representative of the CubeSat orbital environment, with and without magnetic shielding, against the same validated methodology (Sections 9–10).

4 The CT100 Sensor

4.1 Device Overview

The CT100LW-FS6 Tunnel Magnetoresistance (TMR) sensor was selected primarily due to its thermal stability, differential output, small footprint, and cost. The device featured an operating temperature range of $-40\,^{\circ}\text{C}$ to $150\,^{\circ}\text{C}$, making it highly suitable for thermal cycling tests relevant to my partner Daniels Tieu's work. Also, its differential output architecture provided high Common-Mode Rejection Ratio (CMRR), mitigating common-mode noise across its magnetic dynamic range of ± 50 mT with a low linearity error of $\pm 0.5\%$ (between -20 mT and $+20$ mT).

Integrated TMR sensor ICs were explored but ultimately were decided against as raw signal behavior of the sensor was desired, discrete analog sensor enables direct characterization and isolated testing of the TMR elements itself and provided additional flexibility for system integration if needed. Furthermore, its SOT-23-6 package minimizes both PCB footprint and mass, aligning with Size, Weight, and Power (SWaP) constraints typical of CubeSat applications. Sourced at a unit cost of $0.92 via DigiKey (as of April 2026), it provided a highly economical solution for our testing.

The CT100LW-FS6 supports a supply voltage range of 1.0 V to 5.5 V. Because the sensor's differential output sensitivity scales proportionally with supply voltage ($S \propto V_{\text{DD}}$), operating at a higher V_{DD} increases the full-scale differential output voltage swing per unit magnetic field (mV/T). This larger voltage swing enhances the signal-to-noise ratio (SNR). However, a supply voltage of $V_{\text{DD}} = 3.30\,\text{V}$ was selected to match the 3.30 V operating voltage of a STM32 microcontroller. This eliminates the need for additional voltage regulators or signal conditioning level-shifters, simplifying the power electronics if system integration was required.

4.2 Physical Operating Principle

The CT100LW-FS6, like other TMR devices is built around magnetic tunnel junctions (MTJs): a pinned ferromagnetic reference layer, an ultra-thin (~2 nm) insulating crystalline magnesium oxide (MgO) layer and a magnetically free ferromagnetic layer.

Conduction through the junction is done under the quantum tunnelling property, in which electrons have a nonzero probability of passing through the MgO barrier. Figure 1 denotes the MTJs equivalent circuit model.

The basic operation of the MTJ comes down to how easily electrons can pass through the MgO barrier placed between two ferromagnetic layers. The bottom layer serves as a fixed reference with a locked magnetic direction, while the top free layer can rotate its magnetic orientation 360° to align an external magnetic field.

When an external field aligns the free layer's direction (0°) with the bottom reference layer, electrons hop across the insulating barrier much more easily. Higher electron movement per unit of time represents a greater amount of electrical charge ΔQ passing through the barrier

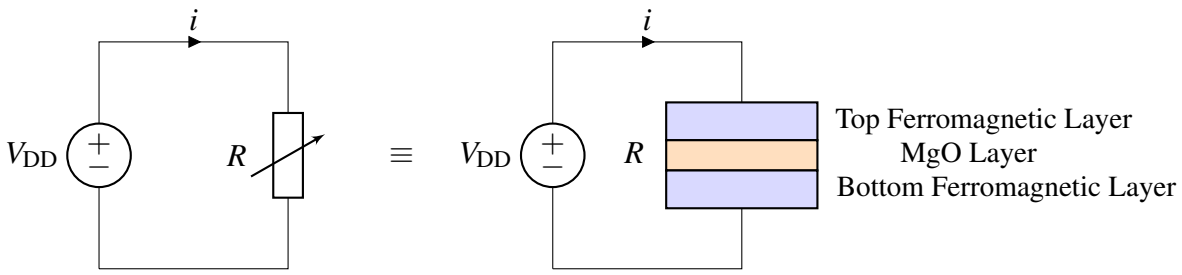


Figure 1: Equivalent electrical circuit model equating a standard variable resistor (left) to the physical three-layer Magnetic Tunnel Junction stack structure (right).

over time Δt. By definition, current is the rate of charge flow per unit of time:

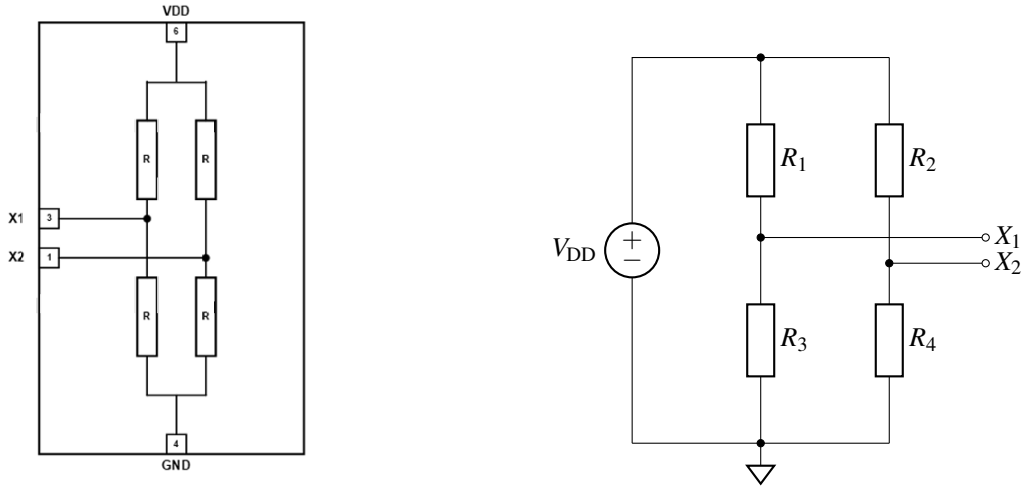
$$I = \frac{\Delta Q}{\Delta t} \tag{1}$$

Therefore, this increased charge movement results in a higher current flow I for a constant supply voltage V_{DD}, putting the sensor into its parallel, lowest resistance state $R = \frac{V_{DD}}{I}$. Conversely, when the field lines points in the opposite direction (180°), fewer electrons can cross, restricting current flow and increasing resistance. If the external magnetic field is applied orthogonal to the fixed reference direction, the sensor cannot detect it, making the element effectively blind to perpendicular fields. Hence, a TMR element can be modelled as a variable resister where it's resistance varies dependant on the polarity and strength of an external magnetic field.

fix this othongal thing just means we don't know what axis the field is more towards parlell or anti parell kinda thing

4.3 Bridge Topology, Output Convention, and Saturation Ceiling

The CT100LW-FS6 (I will refer to the device as the CT100) arranges four such TMR elements in a passive Wheatstone bridge configuration as seen in Figure 2.



(a) CT100 internal Wheatstone-bridge.

(b) Equivalent bridge schematic.

Figure 2: Internal CT100 IC pin configuration alongside its labeled equivalent electrical schematic.

To predict the maximum differential output voltage prior to saturation, the CT100's internal TMR Wheatstone bridge (Figure 2a) was evaluated theoretically. An equivalent schematic with explicitly labeled internal elements is provided in Figure 2b to aid in following the derivation below. According to the device datasheet, the per-element resistance ranges from $R_{\min} \approx 20\,\text{k}\Omega$ where the element experiences parallel magnetic alignment, lowest-resistance state) to $R_{\max} \approx 40\,\text{k}\Omega$ (antiparallel magnetic alignment, highest-resistance state).

As a result of this Wheatstone-bridge configuration, the largest change in $|V_{\text{out}}|$ is $\frac{V_{\text{DD}}}{3}$:

$$|\Delta V_{\text{out,max}}| = \frac{V_{\text{DD}}}{3} \tag{2}$$

This maximum output swing is bounded by each of the internal TMR element resistance range ($R_{\min} \approx 20\,\text{k}\Omega$ to $R_{\max} \approx 40\,\text{k}\Omega$).

The voltage potentials at the two central output nodes, X_1 and X_2, as seen in Figure 2b are determined by the independent voltage divider ratios along the bridge:

$$X_1 = V_{\text{DD}} \cdot \left(\frac{R_3}{R_1 + R_3} \right), \qquad X_2 = V_{\text{DD}} \cdot \left(\frac{R_4}{R_2 + R_4} \right) \tag{3}$$

The general differential output voltage V_{out} across nodes X_1 and X_2 is expressed as:

$$V_{\text{out}} = X_1 - X_2 = V_{\text{DD}} \left(\frac{R_3}{R_1 + R_3} - \frac{R_4}{R_2 + R_4} \right) \tag{4}$$

At $B = 0\,\text{T}$, all four TMR elements exhibit nominal, equal resistance ($R_1 = R_2 = R_3 = R_4$), yielding balanced potentials $X_1 = X_2 = \frac{V_{\text{DD}}}{2}$ and $V_{\text{out}} = 0\,\text{V}$.

A key operational characteristic of the CT100 is that an external magnetic field applied along its sensitive axis drives opposing element pairs in opposite resistance directions. As one pair (R_1, R_4) decreases toward $R_{\min}$, the complementary pair (R_2, R_3) increases toward $R_{\max}$ or vice versa. This response is governed by the orientation of the pinned ferromagnetic reference layer within each TMR element: R_1 and R_4 share an identical reference layer alignment, whereas R_2 and R_3 are pinned in the opposite direction.

At full magnetic saturation, the opposing resistor pairs settle at either $R_{\min}$ and $R_{\max}$. For this derivation, an external magnetic field polarity is assumed such that:

$$R_1 = R_4 = R_{\min}, \quad \text{and} \quad R_2 = R_3 = R_{\max} \tag{5}$$

Substituting these saturation values into Equation 3 gives the node potentials at full saturation:

$$X_{1,\text{sat}} = V_{\text{DD}} \cdot \left(\frac{R_{\max}}{R_{\min} + R_{\max}} \right), \qquad X_{2,\text{sat}} = V_{\text{DD}} \cdot \left(\frac{R_{\min}}{R_{\max} + R_{\min}} \right) \tag{6}$$

Subtracting $X_{2,\text{sat}}$ from $X_{1,\text{sat}}$ yields the maximum theoretical differential output voltage $V_{\text{out,sat}}$:

$$V_{\text{out,sat}} = X_{1,\text{sat}} - X_{2,\text{sat}} = V_{\text{DD}} \left(\frac{R_{\max} - R_{\min}}{R_{\max} + R_{\min}} \right) \tag{7}$$

Substituting the CT100 datasheet resistance values ($R_{\min} = 20\,\text{k}\Omega$ and $R_{\max} = 40\,\text{k}\Omega$):

$$V_{\text{out,sat}} = V_{\text{DD}} \left(\frac{40\,\text{k}\Omega - 20\,\text{k}\Omega}{40\,\text{k}\Omega + 20\,\text{k}\Omega} \right) = V_{\text{DD}} \left(\frac{20}{60} \right) = \frac{V_{\text{DD}}}{3} \tag{8}$$

This derivation proves that the maximum differential output voltage is bound to $|\Delta V_{\text{out,max}}| = \frac{V_{\text{DD}}}{3}$. For a supply voltage of $V_{\text{DD}} = 3.30\,\text{V}$, the theoretical saturation output $|V_{\text{out,sat}}|$ is $1.10\,\text{V}$. This is experimentally validated in Section 6.3.

Furthermore, from this point forward in this report, the sensor's output voltage is defined as $V_{\textbf{out}} = X_1 - X_2$, which was applied across all DMM, oscilloscope, and MCU measurement in this work.

5 Available Instrumentation

Characterization and bench-top tests were conducted using the following laboratory equipment to ensure power delivery and take measurements.

- **Digital Oscilloscope:** Rohde & Schwarz RTB2002 ($2.5\,\text{GS}/\text{s}$ sampling rate, 10-bit A/D converter) for transient analysis, step-response monitoring across nodes X_1 and X_2.
- **Digital Multimeter (DMM):** Keithley 2110 $5\frac{1}{2}$ Digit Dual-Display Multimeter for DC Voltage and Current logging and cross-validating MCU reading accuracy.
- **DC Power Supply Unit (PSU):** Topward dual-tracking linear DC power supply capable of supplying up to $6.0\,\text{A}$ continuously, providing supply voltage and current excitation through the PCB test trace.

6 Standalone Sensor Board

6.1 PCB Design

A standalone sensor board was designed in Altium Designer, the assembled boards were populated with just the CT100 sensor and a single $0.1\,\mu\text{F}$ decoupling capacitor.

The board is driven directly from a laboratory power supply, with the differential outputs (X_1, X_2) measured by a digital oscilloscope and digital multimeter (DMM). Signal amplification was deliberately excluded from this characterization topology. Although raw bridge differential voltages can be relatively small, avoiding amplifier stages eliminates amplifier constraint such as slew rate and bandwidth obscuring the true recovery time of the TMR bridge itself. Probing the unconditioned Wheatstone output directly establishes a true baseline of the sensor's native sensitivity and saturation recovery. This design was also used in Daniel Tieu's work on Thermal Cycling. The PCB can be seen in Figure 3.

6.2 Saturation Testing Methodology

For this board, saturation was induced using an external Neodymium magnet ($> 110\,\text{mT}$), manually brought toward and away from the sensor. This was the simplest available stimulus for an initial characterisation but is not repeatable or automatable.

The Neodymium magnet was manually brought into close proximity along the sensor's to drive the internal TMR bridge elements into parallel/antiparallel saturation. The differential output voltage was monitored using a oscilloscope to confirm that it hit the saturation limit of

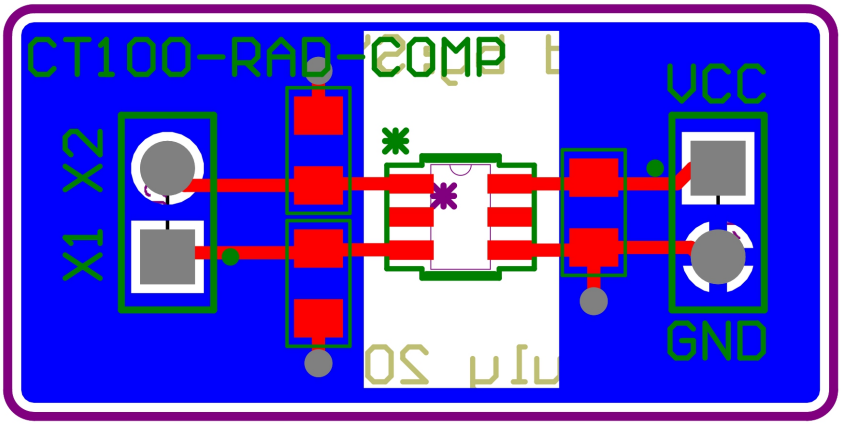


Figure 3: Standalone Sensor PCB

$|\Delta V_{\text{out,max}}| \approx V_{\text{CC}}/3$ derived in Section 4.3. Once the output voltage was confirmed to be at saturation the magnet was rapidly removed, we would then capture the time required for the output to settle back to its steady-state voltage under no external magnetic field excitation.

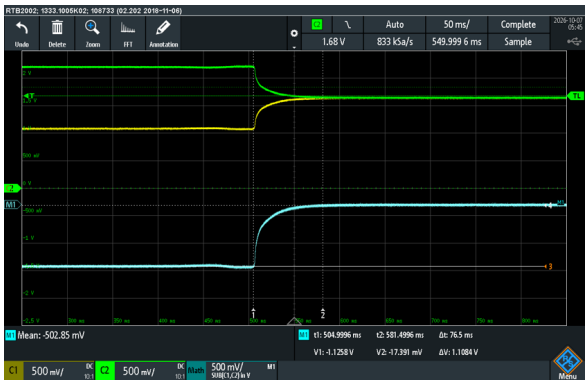
6.3 Sensor-Only Saturation and Recovery Behaviour

Two non-irradiated (Samples 1–2) were driven to saturation each driven to saturation in both field directions, driving both directions checks a specific prediction of the bridge model the saturation ceiling should be symmetric, $\pm V_{\text{CC}}/3$, since the same spin-tunnelling limit applies to either polarity. Table 5 summarises all four trials.

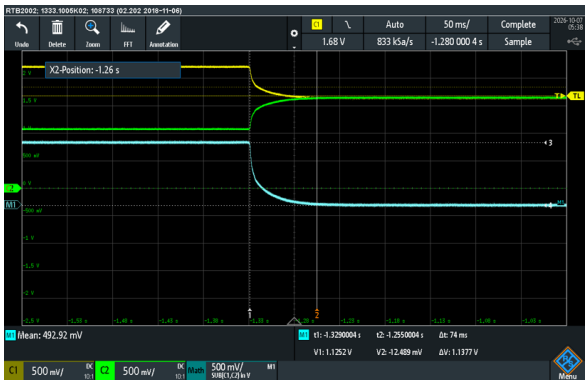
Table 1: Saturation ceiling and recovery time, two samples

Sample	Ceiling (V)	Baseline (mV)	Swing ΔV (V)	Recovery (ms)
1	-1.126	-17.4	1.108	76.5
1	$+1.125$	-12.5	1.138	74.0
2	-1.145	-17.4	1.128	76.0
2	$+1.125$	-12.5	1.138	67.5

To visually document the transient response across both polarities of saturation recovary, Figure 4 presents the oscilloscope captures for Samples 1 and 2. The traces highlight the response as the differential output transitions from the saturated ceiling back to the steady-state baseline once the Neodymium magnet is removed. The blue waveform (M_1) represents the differential output voltage ($V_{\text{out}} = X_1 - X_2$), while the yellow ($C_1$) and green ($C_2$) waveforms represent the individual X_1 and X_2 bridge terminal voltages, respectively.



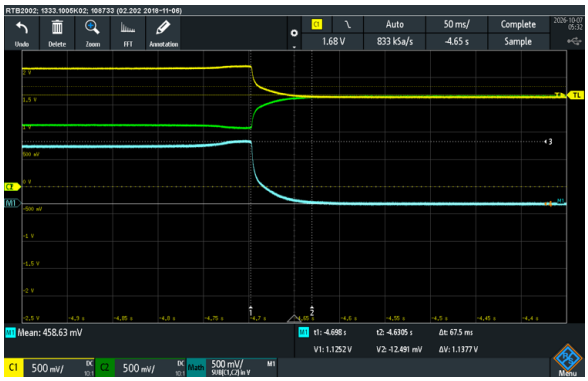
(a) Sample 1: Negative Saturation Recovery



(b) Sample 1: Positive Saturation Recovery



(c) Sample 2: Negative Saturation Recovery



(d) Sample 2: Positive Saturation Recovery

Figure 4: Oscilloscope traces capturing the saturation ceiling and subsequent transient recovery for the non-irradiated sensors.

Across the four trials, the mean transient recovery time following abrupt field removal was $t_{\text{rec}} \approx 73.5\,\text{ms}$, with a maximum recovery time $t_{\text{rec,max}} \approx 76.5\,\text{ms}$. To evaluate the operational significance of this recovery trajectory for spaceflight applications, this response time must be contextualized against typical CubeSat On-Board Computer (OBC) and Attitude Determination and Control System sampling time.

[TODO: add more sources for this shit]

In small satellite architectures, secondary magnetometers used for coarse attitude determination or payload monitoring are typically polled by the OBC rates between 1.0 Hz and 10.0 Hz. For a magnetometer with an update rate of f_s, consecutive readings are spaced $T_s = 1/f_s$ apart. At a 10 Hz update rate specified for a representative CubeSat magnetometer [8], $T_s = 100\,\text{ms}$. Comparing this directly against the measured recovery time (Table 5) gives a margin of only 26.5 ms (26% of T_s) using the mean recovery time (73.5 ms), narrowing to 15 ms (15%) against the single worst trial observed (85 ms).

$$f_{\text{rec,min}} = \frac{1}{t_{\text{rec,max}}} = \frac{1}{76.5\,\text{ms}} \approx 13.07\,\text{Hz} > 10.0\,\text{Hz} \tag{9}$$

This margin is thinner than it first appears, and a simple comparison of recovery *frequency* against sampling *frequency* is not sufficient to assess it. A magnetorquer firing may not be synchronised to the magnetometer's sampling clock, so it may occur at any point within a sampling interval. In the worst-case alignment, a magnetorquer deactivates immediately prior

to a scheduled ADC sample tick. Under this condition, the sensor is sampled while under saturation recovary regardless of whether $f_{\text{rec,min}} > f_s$. Processing an unrecovered transient reading introduces an artificial impulse into onboard state estimators (e.g., Kalman Filters), which can trigger over corrections in attitude or control loop instability.

Therefore, the critical design metric is establishing a time-domain guard window that guarantees complete baseline restoration with margin. Applying a standard $1.5\times$ to $2.0\times$ engineering safety factor to the worst-case observed recovery duration ($t_{\text{rec,max}} = 85.0\,\text{ms}$) defines a recommended guard window of 127.5 ms to 170.0 ms:

$$T_{\text{guard}} = (1.5 \text{ to } 2.0) \times t_{\text{rec,max}} \approx 127.5\,\text{ms to } 170.0\,\text{ms} \tag{10}$$

Because T_{guard} exceeds the single-frame sampling period ($T_s = 100\,\text{ms}$), the flight software should not rely on transient recovery outrunning the sample clock. Instead, the OBC ADCS task must implement an explicit sample-blanking protocol, automatically discarding the first one to two magnetometer samples ($1\text{–}2 \times T_s$) following any magnetorquer firing event.

This is a fixed guard window, not a recalibration: Section 8 tests separately whether saturation itself leaves a lasting zero-point shift requiring recalibration, independent of this transient-recovery guard-window question. Furthermore, this measure of t_{rec} may not be a robust measure as my hand is removing the filed excication its not an automated system the sensor may actual response much quicker so another more definitive way of testing should be explored.

7 Integrated Board: Field Excitation and Design

7.1 Integrated Board Design

To accurately characterize the sensor's sensitivity without relying on a handheld neodymium magnet, a repeatable and quantifiable test field was required.

A current-carrying PCB trace routed directly beneath the sensor die was selected. This DC current induces a magnetic field around the trace that acts upon the TMR sensor to generate a measurable voltage output. It's generated magnetic field is calculated analytically and simulated in Ansys Maxwell, serving as a theoretical benchmark for expected sensor output against practical results. The trace current can be swept up to the laboratory PSU's 6.0 A limit.

While the upper limit of the PSU capability 6.0 A does not generate a strong enough magnetic field to drive the CT100 into saturation, it provides a linear sweep to determine the sensor's sensitivity. Establishing this sensitivity benchmark is critical as later in the characterization, this baseline allows us to measure any resulting hysteresis or degradation in linearity under saturation and/or radiation.

The system architecture of the integrated board revolves around three core components:

- **Microcontroller (MCU):** An STM32G431KBT6 was selected for it's high-performance. The CT100 differential output pair feeds the MCU's internal differential ADC channels directly, sampled at 1.0 kHz. The MCU features a 12-bit ADC and auto-calibration routines were run to minimise discrepancies between ADC's on MCU on different boards.
- **Current Monitor (INA219):** Because the PSU's analog display was not trusted as precise means of determining current, a dedicated INA current-sense with a Kelvin-connected

5.0 mΩ shunt resistor. This was used to back-calculate the magnetic field generated by the trace current acting on the TMR sensor.

- **Low-Dropout (LDO) regulator:** The board is powered via a 5.0 V input rail which is fed into into a LDO regulator to provide a clean 3.30 V supply to both the sensor and MCU. This LDO also provides a high Power Supply Rejection Ratio (PSRR) and sets the subsequent resolution of the TMR sensor.

As shown in Figure 5, the current carrying trace is routed on the bottom layer indicated by the blue polygon with a 2512 shunt resister in series. Figure 6's schematic denotes the positive current ($+i$) sign convention adopted for all subsequent field calculations and experimental sweeps. The Altium schematic for the integrated design can be found in the Appendix A.

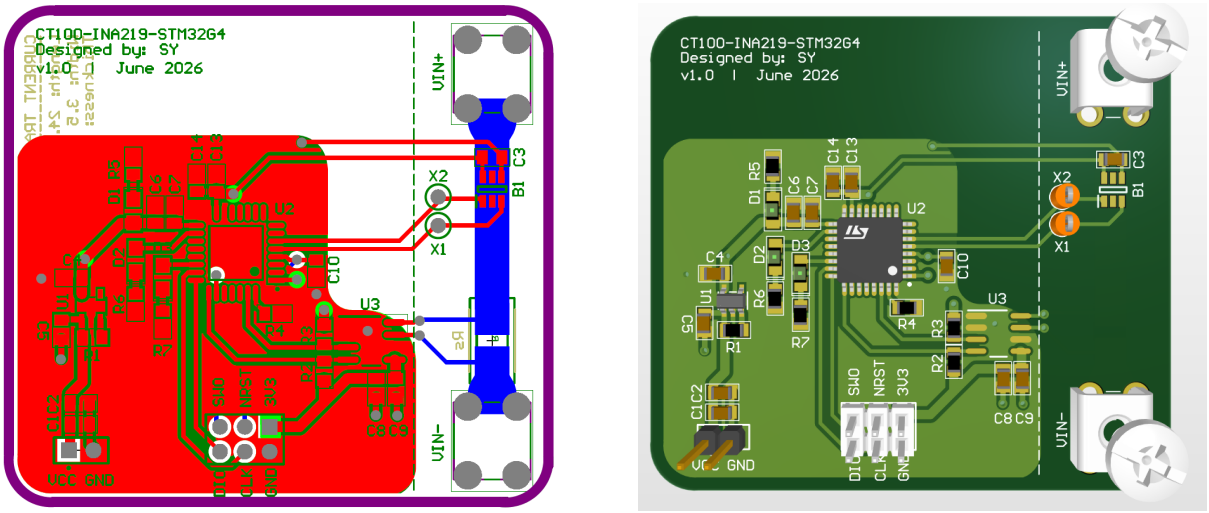


Figure 5: Integrated board layout (left) and rendered assembly view (right).

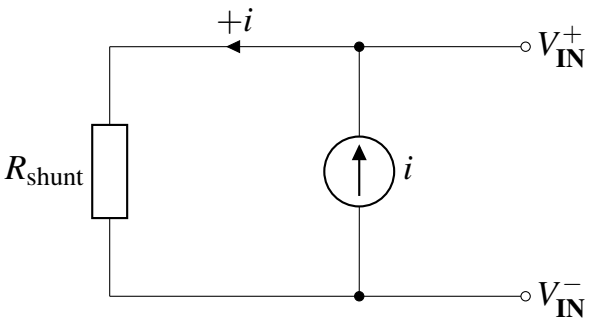


Figure 6: Field excitation circuit schematic and positive trace current convention ($+i$).

Note that while Figure 6's models the excitation as an ideal current source i, the physical setup employs a benchtop PSU operated in Constant Current (CC) mode with the current limit determining the current through R_{shunt} and the trace.

7.2 PCB Trace Design

The current-carrying trace was routed on a four-layer PCB, with layer thicknesses and dielectric parameters detailed in Table 2. The trace was routed such that current flow produces a field

acting on the sensor's most senstive axis, where no copper planes placed near the TMR device and the current trace.

The layout strategy aligns with standard magnetic sensor design like those including Hall Effect sensors. Avoiding copper planes directly beneath and adjacent to the sensor die ensures better magnetic field uniformity and prevents adjacent copper structures from unpredictably altering the local field geometry.**[TODO: add source to back it up]**

Table 2: Four-layer PCB stack-up specification.

Layer	Material	Copper Weight	Thickness
Top Solder	Solder Mask (SM-001)	—	0.0254 mm
Top Layer	Signal Conductor (CF-004)	1.0 oz	0.0350 mm
Layer 2 (GND)	Signal Conductor (CF-001)	0.25 oz	0.0090 mm
Layer 3 (PWR)	Signal Conductor (CF-001)	0.25 oz	0.0090 mm
Bottom Layer	Signal Conductor (CF-004)	1.0 oz	0.0350 mm
Bottom Solder	Solder Mask (SM-001)	—	0.0254 mm

This stack-up yields a total conductive copper thickness of 0.087 99 mm, a dielectric core height of 0.8382 mm, and an overall board thickness of 0.976 99 mm. The excitation trace is routed on the bottom layer with a track width $w = 3.5\,\text{mm}$, capable of currents up to the PSU's 6.0 A limit.

7.3 Analytical Field Model

The horizontal magnetic field at the TMR die height ($d = 0.4229\,\text{mm}$) beneath the centre of the trace ($w = 3.5\,\text{mm}$) is modelled using the standard planar-strip-conductor formulation:

$$B_{\text{planar}} = \mu_0 \left[\frac{I}{\pi w} \arctan\left(\frac{w}{2d}\right) \right]. \tag{11}$$

Normalizing Equation (11) by the trace current I yields the geometric field coupling coefficient k_B:

$$k_B = \frac{B_{\text{planar}}}{I} = \frac{\mu_0}{\pi w} \arctan\left(\frac{w}{2d}\right) \approx 0.1108\,\text{mT A}^{-1}. \tag{12}$$

Evaluated at a representative excitation current of $I = 4.175\,\text{A}$, this model yields a theoretical flux density of 462.70 µT. Combining this field response coefficient k_B with the CT100 sensor's intrinsic sensitivity S_V (V mT^{-1}) gives the expected differential voltage change per unit trace current:

$$\frac{\Delta V_{\text{out}}}{\Delta I} = k_B \cdot S_V. \tag{13}$$

This analytical transfer ratio serves as the theoretical benchmark against which both the Ansys Maxwell finite-element simulations and empirical linear current sweeps are validated.

7.4 Simulated vs. Experimental Field Validation

The analytical field model was cross-checked against a finite-element simulation in Ansys Maxwell. The simulation resolves the full 3D near-field map around the trace rather than the

simplified 2D geometry assumed by Equation (11) the Ansys model can be seen in Figure 7. The two models align closely the simulation outputted $\approx 500\,\mu T$ simulated at 4.175 A while the analytical solution gave 462.70 µT analytically, the formula appears to be a close approximation for the 3D simulation at this geometry.

An experiment was conducted to compare any discrepancies between our analytical and experimental results. We measured the output voltage at 4.175 A to be $\approx 16\,mV$ using a DMM. Which was roughly double what the simulatation predicted $\approx 7.43\,mV$. This discrepancy, and its possible causes; external ambient field offsets is examined in Section 10.

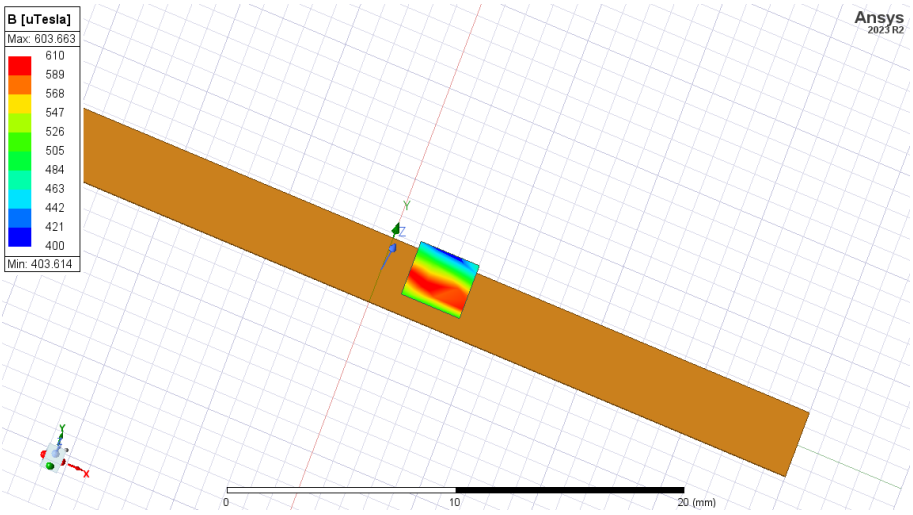


Figure 7: Ansys near-field simulation of the current trace.

[TODO: check math and calcs]

7.5 Firmware

The firmware is built upon the STM32 Hardware Abstraction Layer (HAL). The MCU's ADC is configured in differential-pair mode, sampling $X_1 - X_2$ directly and converting the raw 12-bit value to volts via:

$$X_1 - X_2 = \left(\frac{\mathrm{ADC_{raw}} - 2048}{2048} \right) \times V_{\mathrm{DD}} \tag{14}$$

where 2048 is the mid-scale ADC code corresponding to the zero-differential (balanced-bridge) midpoint. Hardware ADC self-calibration (`HAL_ADCEx_Calibration_Start`) is executed automatically on every boot before any conversions occur. The MCU continuously polls the INA219 over I^2C for trace current and streams both the current and voltage values out in real time over the Serial Wire Viewer (SWV) debug channel for offline analysis in MATLAB. The code can be found in Appendix B.

7.6 Design Note for Future PCB Revisions

Because the sensor bridge is nominally balanced at $V_{\mathrm{DD}}/2$ (Section 4.3), the ADC channel monitoring it spends its full dynamic range on either side of that midpoint. Any measurement taken near the zero-differential code inherits the ADC's own quantization and calibration uncertainty around that center point (see the code-1042 fault discussed in Section **??**).

A future PCB revision could integrate a hardware offset to bias the bridge's reference point to a non-zero value. This is expected to (i) reduce measurement error introduced by operating close to the ADC's most quantization-sensitive midpoint code, and (ii) artificially reduce the external magnetic field magnitude required to drive the sensor to its effective saturation bounds for testing purposes. **[TODO: This is currently a design intention rather than a tested result – expand with the specific biasing approach and expected numbers if it is implemented on a future board revision.]**

8 Non-Irradiated Sensitivity and Saturation Characterisation

8.1 Saturation Recovery on the Integrated Board

On the integrated board, with a regulator input of $V_{CC} = 4.97\,\mathrm{V}$ and a TMR sensor supply of $V_{DD} = 3.32\,\mathrm{V}$.

Table 3: Integrated board, non-irradiated: saturation ceiling, baseline shift and recovery.

Run	Direction	Ceiling (V)	Baseline (mV)		Swing (V)	τ (ms)	Recovery (ms)
			before	after			
A	Negative	−1.152	−6.4	−8.1	1.144	13.1	65.2
B	Positive	+1.138	−8.1	+12.9	1.125	13.3	60.2

(a) Non-irradiated, negative saturation recovery (b) Non-irradiated, positive saturation recovery

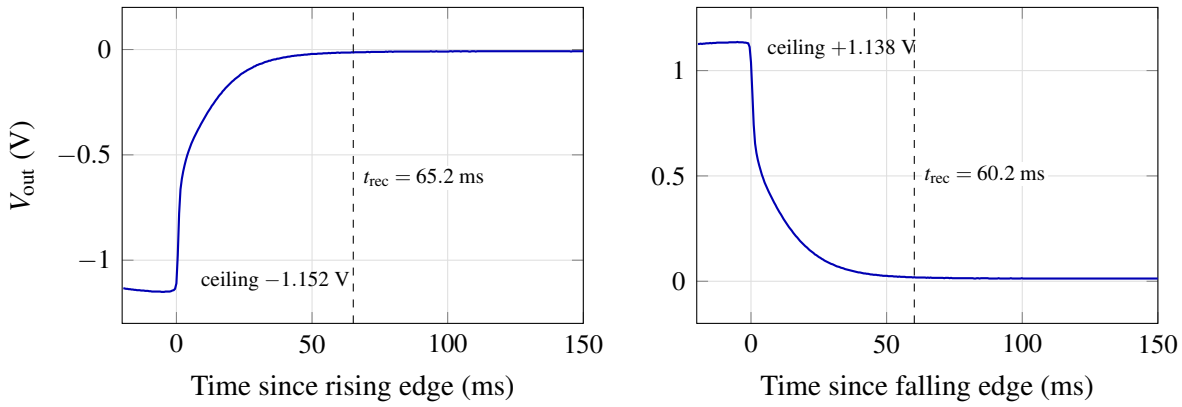


Figure 8: Non-Irradiated Saturation recovery on the integrated board after the magnet is removed, for both field directions.

Comparing the saturation behaviour of the integrated board against the standalone sensor baselines (Section 6.3) reveals highly consistent performance across both hardware configurations. In both test environments, the TMR elements exhibited symmetric saturation limits. The integrated board reached voltage ceilings of $-1.152\,\mathrm{V}$ and $1.138\,\mathrm{V}$, which closely match the $-1.145\,\mathrm{V}$ and $1.125\,\mathrm{V}$ extremes observed in the standalone samples.

The transient recovery times demonstrate tight alignment between the two configurations. The integrated board achieved full baseline restoration slightly faster of $t_{rec} \approx 60.2$–$65.2\,\mathrm{ms}$ than the

standalone sensors $t_{\text{rec}} \approx 67.5$–$76.5\,\text{ms}$. As noted in Section 6.2, the physical removal of the Neodymium magnet was performed manually.

8.2 Current Sweep: Data and Sensitivity Fit

Current-sweep data was logged via the MCU's SWV output, exported, and processed in MATLAB. Rather than relying on single instantaneous readings, each current setting was captured over a 20-second window and averaged. A single-sample readings were found to be unreliable due to the MCU ADC's quantisation behaviour where raw SWV data at a given current setting was observed hopping between a small number of discrete levels spaced at multiples of the 1.61 mV LSB typically spanning 1–2 LSBs of scatter around the true mean. Note that for this non-irradiated integrated board the saturation recovery test was performed before the current-sweep test.

The trace current was swept from -3.0 to $+3.0\,\text{A}$ in approximately $0.5\,\text{A}$ steps, all within a single session at $V_{\text{DD}} = 3.32\,\text{V}$ (Table 4).

Table 4: Non-Irradiated & No Shielding Current Sweep Data

DMM Current (mA)	INA Current (mA)	DMM (mV)	MCU (mV)	Scope (mV)
−2991.0	−2966.7	8.634	8.069	−12.207
−2498.5	−2478.2	9.860	9.571	−10.986
−2029.0	−2012.6	9.950	9.584	−10.376
−1501.0	−1487.7	11.690	11.249	−9.766
−1022.0	−1016.6	11.250	11.107	−9.308
−499.0	−498.2	12.650	12.022	−10.681
−0.7	−6.3	12.980	12.847	−8.594
499.0	487.8	14.450	13.628	−7.477
1022.0	1005.2	14.210	13.796	−5.035
1501.0	1477.2	15.697	14.790	−6.104
2029.0	2002.1	15.720	15.020	−3.516
2498.5	2466.4	16.500	16.068	−3.906
2991.0	2954.8	16.570	16.088	−5.030

A linear fit of bridge output against trace current gives a sensitivity of $1.356\,\text{mV}\,\text{A}^{-1}$ on the DMM and $1.315\,\text{mV}\,\text{A}^{-1}$ on the MCU.

The CT100 datasheet specifies a typical full-bridge sensitivity of $4.5\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}$, with a specified range of $3.8\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}$ to $5.2\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}$. At the 3.32 V supply voltage used for this characterisation, the corresponding theoretical voltage sensitivity is therefore

$$S_{V,\text{theory}} = S_{\text{CT100}} V_{\text{DD}} = \left(4.5\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}\right)(3.32\,\text{V}) = 14.94\,\text{mV}\,\text{mT}^{-1}. \tag{15}$$

The analytical field model in Equation (11) gives a field coefficient of approximately $0.1108\,\text{mT}\,\text{A}^{-1}$ for the trace geometry used in the test. Combining this field response with the CT100 sensitivity gives the expected voltage change per unit trace current:

$$\frac{\Delta V_{\text{out}}}{\Delta I} = S_{V,\text{theory}} \frac{\Delta B}{\Delta I} = \left(14.94\,\text{mV}\,\text{mT}^{-1}\right)\left(0.1108\,\text{mT}\,\text{A}^{-1}\right) = 1.655\,\text{mV}\,\text{A}^{-1}. \tag{16}$$

Thus, for a given trace current, the theoretical sensor output can be written as

$$V_{\text{out,theory}} = V_0 + \left(1.655\,\text{mV}\,\text{A}^{-1}\right) I, \tag{17}$$

where V_0 is the zero-field output voltage. Figure 9 below uses a V_0 value of 12.980 mV to visually compare the voltage *slope*; it is not a theoretical prediction of the CT100 offset voltage. Figure 10 plots how close the INA device follows the DMM measurement.

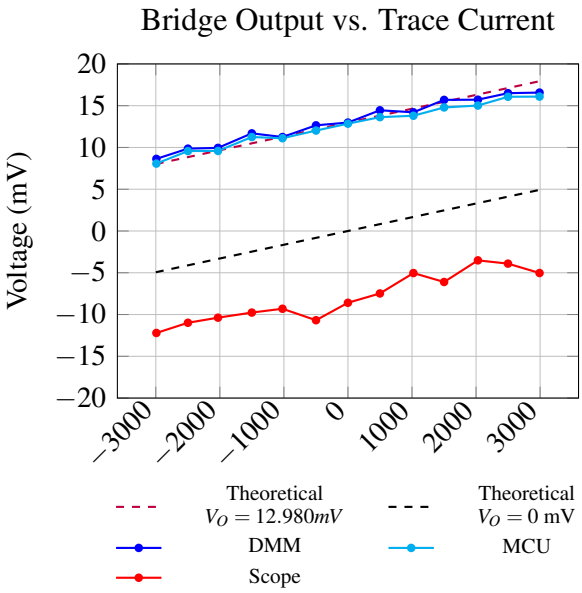


Figure 9: Voltage cross-validation during the non-irradiated, unshielded current sweep. The dashed lines shows the theoretical CT100 responses.

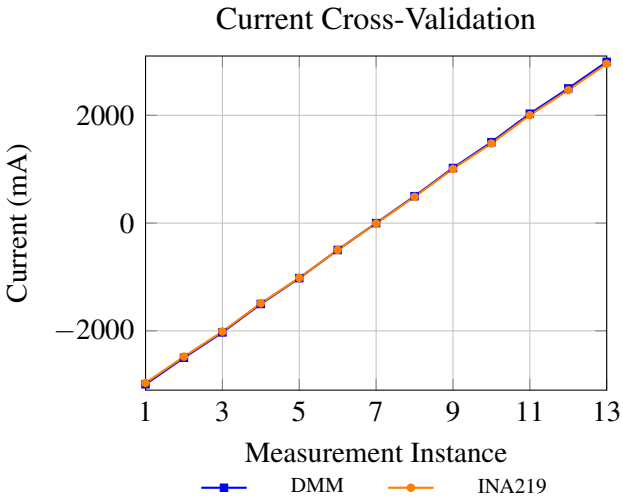


Figure 10: Current cross-validation across the 13-point sweep.

The theoretical current-to-voltage slope of 1.645 mV A^{-1} is greater than both measured slopes. For the DMM, the measured slope is approximately 82.4% of the theoretical value, corresponding to a sensitivity deficit of approximately 17.6%. The MCU slope is approximately 79.9% of the theoretical value. The close agreement between the two instruments and their similar deviation from the theoretical slope indicate that the difference is not primarily an instrumentation-specific offset. The DMM and MCU measurements closely track one another, while the oscilloscope exhibits a different current-dependent response.

The current-measurement cross-validation showed that the INA219 tracked the DMM closely across the full current sweep, with a near-unity linear relationship and a maximum difference of only 36.2 mA. The INA219 therefore provides sufficient accuracy for the current checks required in the subsequent testing. As the INA219 is already integrated into the measurement system, it will be used as the primary current measurement for future sweeps. This removes the need to repeat each current measurement using the external DMM and reduces the time required for data collection while maintaining a validated measurement reference.

The oscilloscope measurements were not consistent with the DMM and MCU voltage measurements during the DC current sweep, showing a substantially different response and negative output values. Since the DMM and MCU measurements agreed closely with one another, while the oscilloscope response differed significantly. Consequently, oscilloscope measurements will

not be used for subsequent DC voltage or sensitivity checks. The oscilloscope will instead be reserved for transient measurements, where its time-domain capability provides information that cannot be obtained from the DMM.

Once driven into saturation, the sensor was observed to return to a *new*, offset baseline rather than the original zero-field value, with the output remaining linear about this new offset for subsequent field excursions. This is consistent with magnetic hysteresis in the free layer: after saturation, the free layer's magnetisation does not fully relax back to its original zero-field orientation, leaving a small permanent shift in the balance point of the bridge. **[TODO: Quantify the magnitude of this offset (mV), confirm whether it is stable over time/repeated cycles, and clarify which board/session this was observed on, since it appears to sit alongside an earlier record of the output returning "instantly" to the original linear baseline after saturation – these two observations should be reconciled (e.g. the earlier note may have predated the sensor's first-ever saturation, with hysteresis only appearing from the first saturation event onward). This has a direct design implication aligns with current research state it and talk about implications of why it would be bad.]**

9 Radiation Testing (Total Ionising Dose)

9.1 Rationale and Test Isolation Strategy

Crystalline MgO, the MTJ's insulating barrier, is expected to be substantially more radiation-tolerant than the amorphous SiO_2 gate oxide used in conventional silicon devices, for the reasons discussed in Section 2.2. To test this without confounding the result with unrelated failure modes, only the passive CT100 sensor itself was irradiated by itself.

9.2 Experimental Setup

Ten CT100 sensors were exposed using a 320 kV tube potential (tungsten target, 30° anode angle, 3 mm beryllium filter) at a beam current of 10 mA. The beam was seasoned (ramped up to full parameters) from 15:55 to 16:05 NZST on 22 July 2026, and held at full 320 kV/10 mA output until switch-off at 19:26 NZST, giving 3 h 21 min (12 060 s) of full-power exposure.

The physical test setup inside the irradiation chamber is shown in Figure 11. The dose rate at the sensor location was calculated as 0.0338 $\mathrm{Gy\,s^{-1}}$ (Si), equal to 3.38 rad(Si)/s, based on the beam path approximated as 315 mm of air followed by 5 mm of aluminium attenuation before reaching the TMR sensor, itself modelled as a silicon block for absorption purposes. Applied over the 12 060 s full-power exposure window, this gives a total absorbed dose of 407.6 Gy (40.76 krad(Si)). This dose figure does not include the lower, ramping dose accumulated during the 15:55–16:05 seasoning period, so the total absorbed dose reported here is a conservative estimate.

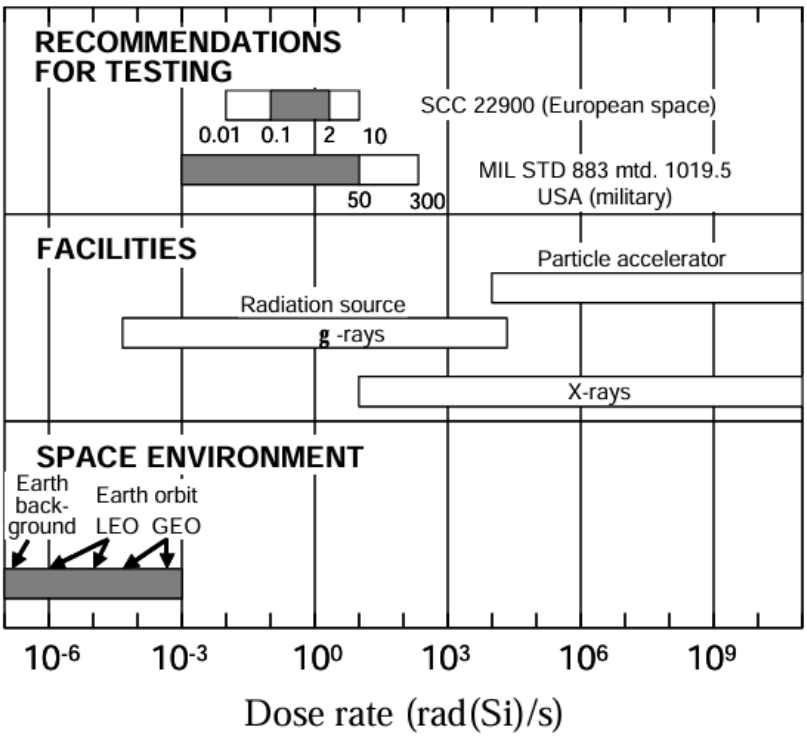


Figure 11: Comparison of the dose rates found in the radiation environments and the rates used in the testing of components for tolerance.

To contextualize the severity and relevance of this exposure, these parameters were evaluated against the radiation hardness assurance framework outlined by Poivey [**?**]. Actual natural space environments, such as Low Earth Orbit (LEO) and Geosynchronous Equatorial Orbit (GEO), typically expose components to extremely low, continuous dose rates below $10^{-3}\,\mathrm{rad(Si)/s}$. The experimental dose rate of $3.38\,\mathrm{rad(Si)/s}$ provides a robust accelerated testing environment that effectively simulates cumulative degradation over a practical laboratory timeframe. Notably, this rate aligns directly with the SCC 22900 (European space) recommendations, which mandate testing dose rates between approximately 0.01 and $10\,\mathrm{rad(Si)/s}$[cite: 21]. Consequently, the accumulated $40.76\,\mathrm{krad(Si)}$ represents a fully compliant and highly relevant benchmark for standard orbital spaceflight qualification.

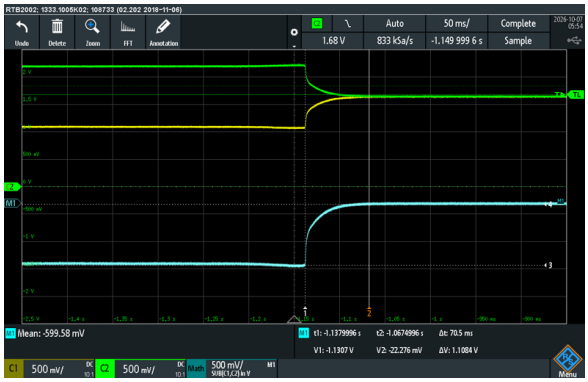
9.3 Sensor-Only Board Saturation Response Post-Irradiation

To evaluate whether radiation exposure alters sensor saturation limits or transient recovery dynamics, two irradiated units (Samples 3 and 4) were driven to magnetic saturation following the methodology established in Section 6.3. Table 5 summarizes the saturation ceilings, baseline offsets, differential swings, and recovery times across all eight trials for both non-irradiated and irradiated sample groups for comparison.

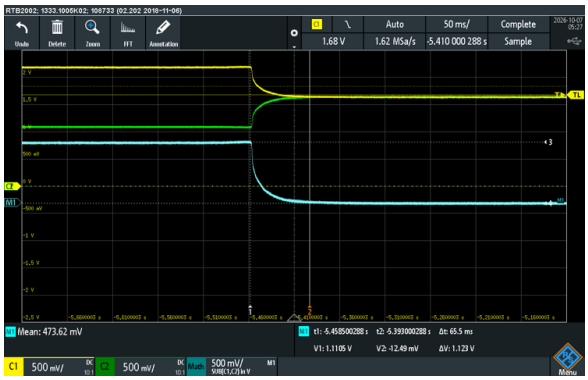
Figure 12 shows the oscilloscope captures for the irradiated samples. Following the same trace colour assignment used in the non-irradiated characterization, the blue waveform (M_1) represents the differential output voltage ($V_{\mathrm{out}} = X_1 - X_2$), while the yellow ($C_1$) and green ($C_2$) waveforms represent the individual X_1 and X_2 bridge terminal voltages, respectively.

Table 5: Saturation ceiling and recovery time, two irradiated samples.

Sample	Ceiling (V)	Baseline (mV)	Swing ΔV (V)	Recovery (ms)
1	−1.126	−17.4	1.108	76.5
1	+1.125	−12.5	1.138	74.0
2	−1.145	−17.4	1.128	76.0
2	+1.125	−12.5	1.138	67.5
3	−1.131	−22.3	1.108	70.5
3	+1.111	−12.5	1.123	65.5
4	−1.145	−22.3	1.123	78.0
4	+1.111	−22.3	1.133	85.0



(a) Sample 3: Negative Saturation Recovery



(b) Sample 3: Positive Saturation Recovery



(c) Sample 4: Negative Saturation Recovery



(d) Sample 4): Positive Saturation Recovery

Figure 12: Oscilloscope captures displaying post-irradiation saturation voltage clamping and transient relaxation trajectories for irradiated sensor board Samples 3 and 4 across both field polarities.

Averaging across both polarities and units within each group, the non-irradiated group exhibits a mean saturation ceiling of 1.130 V compared to 1.124 V for the irradiated group, representing a 0.5% difference. The mean recovery time across groups is 73.5 ms non-irradiated versus 74.75 ms irradiated, a 1.7% difference.

Because the performance difference between the non-irradiated and irradiated groups averages (0.5% on saturation ceiling and 1.7% on recovery duration) falls strictly within the non-irradiated unit-to-unit spreads, the differences observed are attributable to manufacturing vari-

ance rather than total ionizing dose (TID) degradation. Consequently, neither the saturation voltage ceiling nor the transient recovery demonstrate detectable permanent radiation-induced degradation at the time of testing. Note, both irradiated and non-irradiated samples were tested in the same session on the 7th of October 2026.

9.4 Quantitative Sweep on the Integrated Board

Following transfer to a integrated board the same current-sweep experiment as Section 8.2 was repeated results can be found in Table 6. Note, the irradated sensor had not been exposed to saturating magnetic fields prior to irradiation, both pre- and post-irradiation sweeps were performed on a bridge with zero magnetic history.

Table 6: Irradiated sample current sweep.

Current (mA)	DMM (mV)	MCU (mV)
−2999.0	−3.723	−3.258
−2595.0	−3.123	−2.803
−1999.0	−2.570	−1.748
−1512.0	−1.620	−0.044
−1017.0	−1.050	−0.001
−498.0	−0.100	0.463
−0.7	0.269	0.308
498.0	1.130	1.749
1017.0	1.640	2.657
1512.0	2.098	3.178
1999.0	2.720	3.298
2595.0	3.303	4.350
2999.0	3.912	4.834

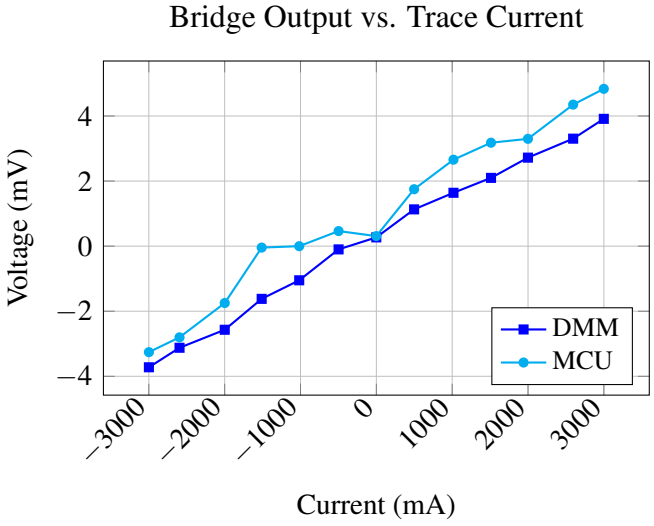


Figure 13: Plot of irradiated DMM and MCU sweeps detailed in Table 6.

The same linear-fit procedure gives a sensitivity of $1.269\,\mathrm{mV\,A^{-1}}$ on the DMM ($R^2 = 0.995$) and $1.312\,\mathrm{mV\,A^{-1}}$ on the MCU ($R^2 = 0.975$), corresponding to 93.6% (DMM) and 99.8% (MCU) of the unirradiated sensitivity (Section 8.2). Both figures are well within the level of session-to-session variation already observed on unirradiated hardware (Section **??**), with no consistent direction of change between the two independent instruments. *No statistically significant change in sensitivity was found at the tested dose.* Given that this specific radiation configuration does not appear elsewhere in the TMR-sensor literature reviewed for this project (Section 2.2, pending confirmation), this result is offered as a preliminary but novel data point for TMR radiation tolerance under this exposure regime, supporting the broader literature consensus on MTJ radiation tolerance (Section 2.2).

9.5 Irradiated Saturation Recovery on the Integrated Board

Following the 40.76 krad(Si) Total Ionizing Dose (TID) exposure, the saturation recovery of the integrated board was evaluated with a regulator input of $V_{CC} = 5.00\,\mathrm{V}$ and a measured TMR sensor supply of $V_{DD} = 3.289\,\mathrm{V}$, the irradiated board was subjected to the same magnetic saturation procedure as in Section 8.1. The resulting recovery waveforms are plotted in Figure 14

and the key metrics are summarized in Table 7. The following irradiated data was collected on the 9th of October 2026.

Table 7: Integrated board, irradiated: saturation ceiling, baseline shift and recovery.

Run	Direction	Ceiling (V)	Baseline (mV)		Swing (V)	τ (ms)	Recovery (ms)
			before	after			
A	Negative	-1.146	0.0	0.0	1.146	13.8	66.7
B	Positive	$+1.149$	0.0	$+1.6$	1.147	12.7	63.2

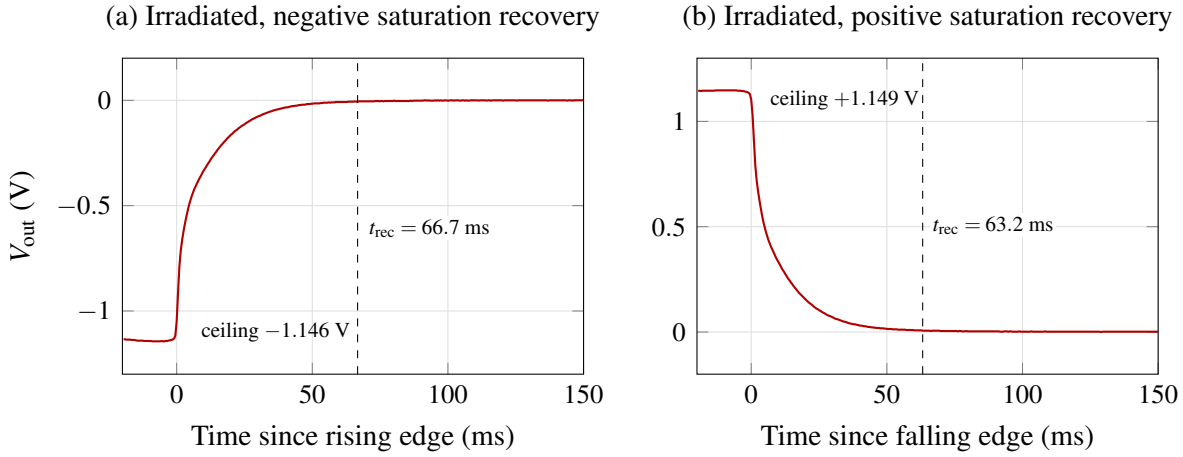


Figure 14: Irradiated Saturation recovery on the integrated board after the magnet is removed, for both field directions.

10 Magnetic Shielding

10.1 Enclosure Design

To address the possible ambient-field discrepancy identified in Section 8.2, a 3D-printed enclosure incorporating ferrite shielding material was designed. The primary mechanism of magnetic shielding relies on enclosing the PCB with a high-permeability material, which provides a low-reluctance path that shunts external magnetic flux away from the device and thus influencing our TMR sensor readings.

Mu-metal, a nickel-iron alloy, is traditionally the gold standard for DC and low-frequency magnetic shielding due to its exceptionally high initial permeability (typically $\mu_i > 80,000$). However, it was not used for this enclosure due to strict cost constraints and its sensitivity to mechanical stress, which requires expensive re-annealing after shaping. **[TODO: add sources and whatever]**

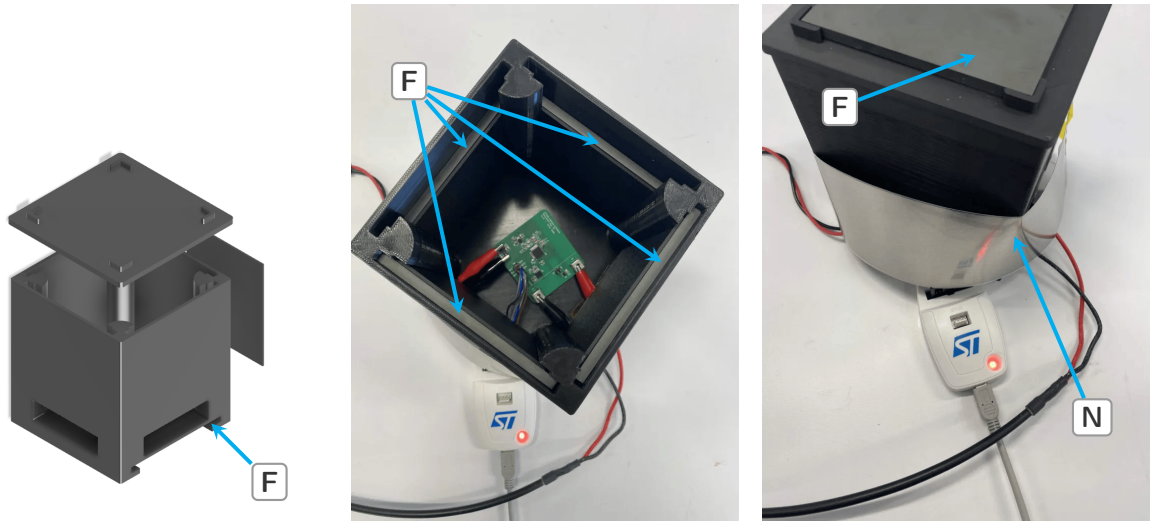
Instead, a MnZn-based SIFERRIT material TDK N95 was selected as a cost-effective preliminary alternative. According to the manufacturer specifications, N95 ferrite offers an initial relative permeability (μ_i) of 3000 at 25°C. While this is an order of magnitude lower than Mu-metal, it should still provides some flux-shunting capability. Six TDK N95 ferrite slabs, each measuring 100 mm × 100 mm × 5 mm, were arranged to construct the shielding walls. With

a saturation flux density B_S of 525 mT at room temperature, the 5 mm the slabs are unlikely to saturation under our testing conditions so our permeability remains linear.

To enable continuous operation and data logging during testing, power and data connections via an ST-Link programmer must be routed to the enclosed PCB. Consequently, a physical cut-out was designed into the enclosures to accommodate this. However, any gap effects the reluctance of the shielding and allows for ambient magnetic flux to leak and impact the TMR readings.

To mitigate this flux leakage, a flexible nanocrystalline shielding foil was used. This highly permeable material was used to cover cable cut-out.

The magnetic shielding progression—including the printed chassis slots, the internal board placement (Figure 15b), and the final exterior wrap (Figure 15c)—can be seen in Figure 15.



(a) Enclosure CAD render. (b) Internal PCB mounting. (c) Exterior shielding layer.

Figure 15: Development of the magnetic shielding enclosure. (a) The 3D-printed chassis. (b) The integrated sensor board mounted inside. (c) The assembled test configuration wrapped in shielding. Annotations indicate the placement of ferrite plates (**F**) and flexible nanocrystalline shielding (**N**).

10.2 Shielded Sweep Results

With MCU-DMM agreement already established across the full current range (Section 8), the shielded current sweep was conducted using MCU-only logging (no DMM or oscilloscope): the earlier cross-validation (Section 8) showed MCU and DMM readings agreeing to within a fraction of an ADC LSB across the unshielded sweep, so introducing the DMM into the shielded enclosure was judged to add setup complexity and added another source of possible stray magnetic field.

The non-irradiated and irradiated integrated boards was tested inside the shielded enclosure under the same sweep procedure as in Section 8. Note that the irradiated TMR sensor tested under this shielding condition had not previously been driven into saturation.

The raw measurement data for both the non-irradiated and irradiated shielded sweeps are detailed in Table 8 and Table 9, respectively. To visualize the effect of the magnetic shielding, Figure 16 compares these shielded sweeps directly against their unshielded counterparts.

Operating conditions are as follows, testing perfomed at a $V_{DD} = 3.29\,V$.

Table 8: Shielded, non-irradiated current sweep (MCU only): bridge output vs. trace current.

Current (mA)	MCU (mV)
−3043.4	8.039
−2522.5	8.549
−2057.7	9.594
−1540.3	9.949
−1040.2	11.170
−546.2	11.319
−6.3	12.705
535.4	12.946
996.2	13.931
1523.0	14.530
2038.1	15.532
2486.1	16.103
2988.7	16.474

Table 9: Irradiated, shielded current sweep (MCU only): bridge output vs. trace current.

Current (mA)	MCU (mV)
−3007.1	−3.287
−2501.2	−2.769
−2054.6	−1.130
−1567.3	0.069
−1037.7	−0.001
−532.1	0.107
−6.3	1.318
512.5	1.699
1011.4	2.837
1557.8	3.241
2048.6	3.933
2560.6	4.807
3025.4	4.932

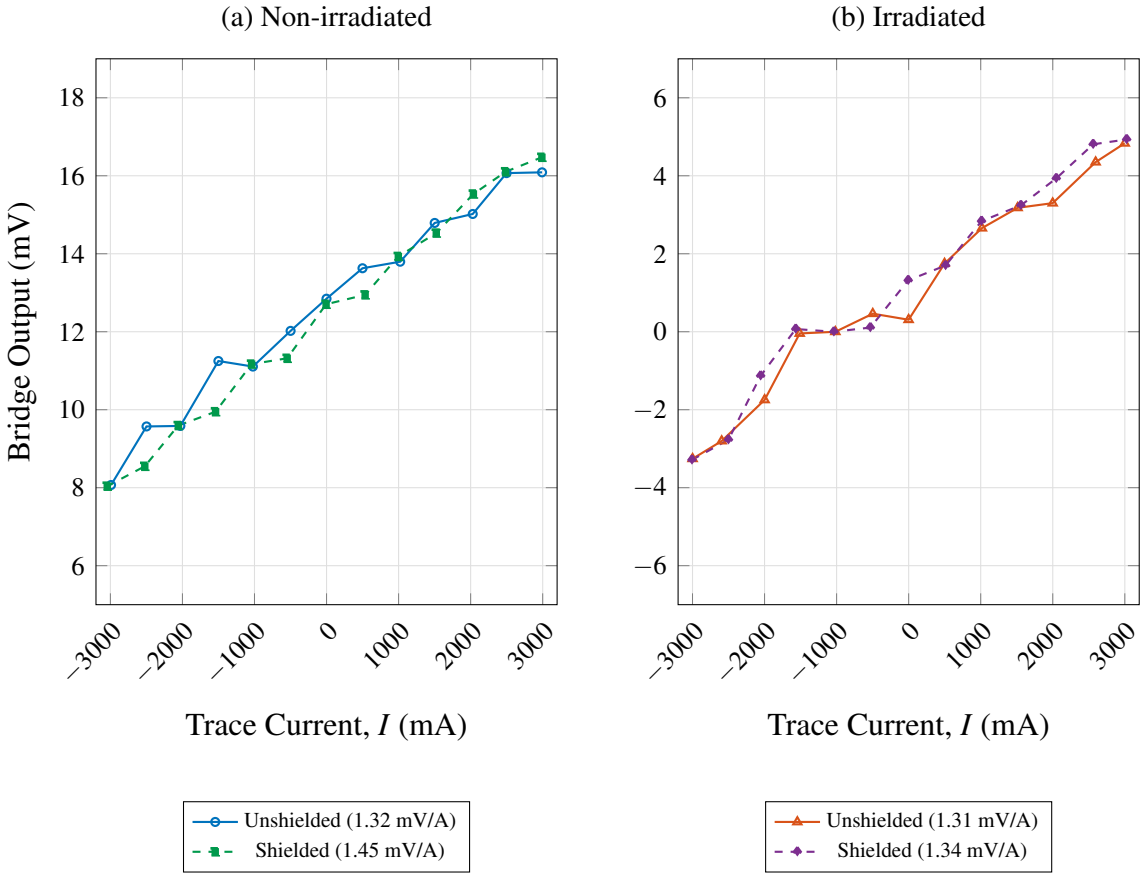


Figure 16: Shielded vs. unshielded current sweeps (MCU readings) for (a) the non-irradiated and (b) the irradiated sample.

Shielding changes the gradient by $+10\%$ on the non-irradiated board and $+2.3\%$ on the irradiated board.

10.3 Session-to-Session Variance

To better quantify the expected measurement tolerance and validate earlier findings, a secondary current sweep was performed on the non-irradiated shielded MCU board on. This validation test was conducted at $V_{DD} = 3.29\,\text{V}$.

The raw data logged for this secondary shielded sweep is detailed in Table 10. Figure 17 visualizes this data alongside the initial shielded non-irradiated sweep to illustrate the session-to-session any offset and sensitivity drift.

Table 10: Validation Sweep: Non-Irradiated Shielded Board

INA Current (mA)	MCU (mV)
−3046.47	8.039
−2492.64	8.595
−2009.76	9.604
−1493.87	10.014
−1037.81	11.110
−507.16	11.312
−4.58	12.514
494.27	12.888
1055.68	13.579
1515.83	14.455
2016.78	14.890
2487.65	15.978
3006.08	16.192

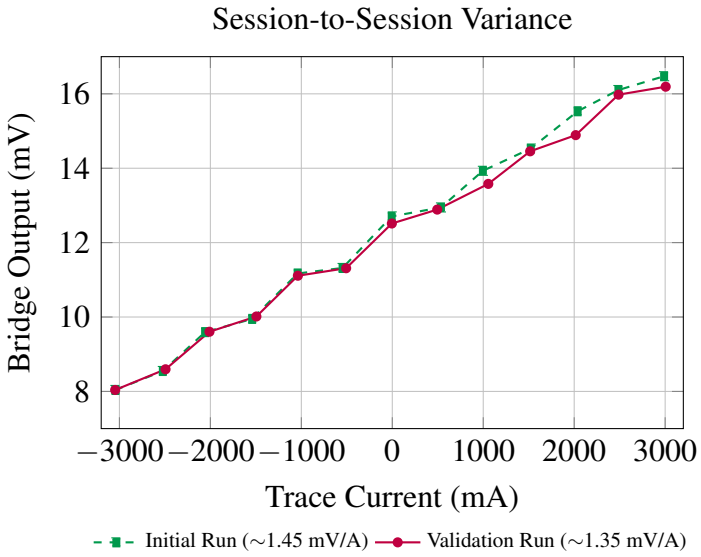


Figure 17: Plots of two discrete testing sessions on the same shielded, non-irradiated board.

A linear estimation of the validation sweep data yields an approximate sensitivity of $1.35\,\text{mV}\,\text{A}^{-1}$. When compared to the initial shielded run's sensitivity of $1.45\,\text{mV}\,\text{A}^{-1}$ (Section 10.2), this reveals an approximate 7% discrepancy attributable to standard session-to-session environmental and setup variance.

Crucially, this baseline session to session variance contextualizes the earlier shielding tests. The 2.3% to 10% sensitivity differences previously observed between unshielded and shielded states fall within this expected session to session tolerance. Consequently, the data does not demonstrate a distinct shielding effect, the observed deviations are likely artifacts of standard testing drift rather than a physical shift in the sensor's ambient response. Furthermore, no statistically significant change in sensitivity is apparent across either shielding condition.

11 Quantitative Magnetic Hysteresis Testing

To isolate the effects of magnetic history from potential board-level hysteresis, an un-irradiated TMR sensor with no prior magnetic exposure was newly soldered onto the integrated board.

Current sweep characterisation was performed on 12 October 2026, operating at $V_{CC} = 5.00\,\text{V}$ and $V_{DD} = 3.321\,\text{V}$.

The board was first swept to establish a pristine baseline. It was then driven into magnetic saturation, followed by a second identical current sweep to observe any permanent baseline shift. The resulting data demonstrates a severe zero-point shift post-saturation, abruptly shifting the baseline by approximately 16 mV, while the overall sensitivity remained largely unaffected.

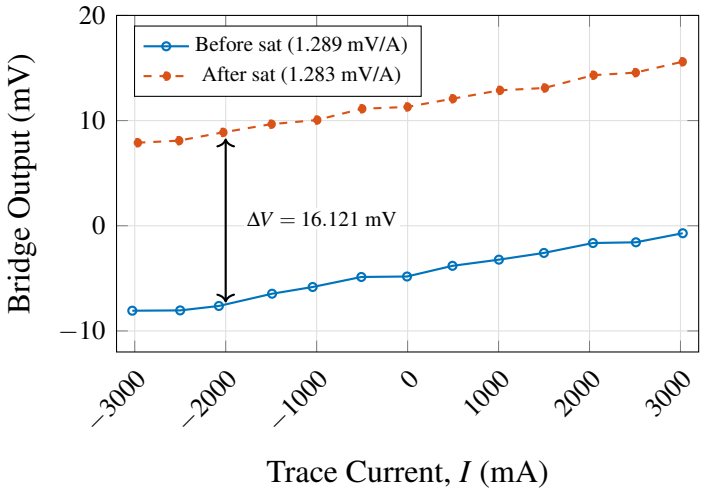
Table 11: Before Saturation

INA (mA)	MCU (mV)
−3026.78	−8.076
−2500.36	−8.047
−2074.76	−7.625
−1490.19	−6.462
−1043.62	−5.821
−508.79	−4.869
−6.30	−4.824
493.28	−3.809
1003.71	−3.222
1498.85	−2.582
2037.68	−1.641
2510.09	−1.571
3025.05	−0.709

Table 12: After Saturation

INA (mA)	MCU (mV)
−2971.20	7.896
−2516.89	8.091
−2031.67	8.878
−1500.70	9.660
−999.23	10.051
−504.66	11.124
−6.15	11.297
492.07	12.077
1010.23	12.875
1504.07	13.100
2039.39	14.316
2502.52	14.558
3018.47	15.590

(a) Raw output



(b) Offset removed

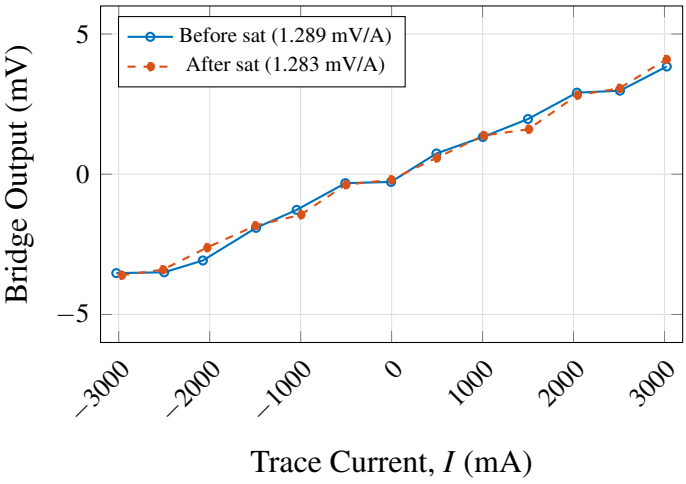


Figure 18: Current sweep on a fresh, never-saturated sensor before (solid) and after (dashed) being driven to saturation. (a) Raw output: the curve moves up by 16.121 mV. (b) With offset removed, sweeps overlap.

The saturation events applied to the fresh sensor reveal a distinct hysteresis in the zero-current baseline, though the device's fundamental sensitivity remains highly stable. As detailed in Table 13, the sensor was sequentially driven into negative and then positive saturation to evaluate its recovery and offset behavior.

Recovery Transients: When the saturating magnetic field is removed, the sensor output decays

Table 13: Fresh sensor: saturation ceiling, baseline shift and recovery (negative direction first, then positive).

Run	Direction	Ceiling (V)	Baseline (mV)		Swing (V)	τ (ms)	Recovery (ms)
			before	after			
A	Negative	−1.107	−4.8	−1.6	1.105	8.6	40.1
B	Positive	+1.117	−1.6	+11.3	1.105	9.6	48.7

from its respective ceiling (-1.107 V for negative saturation, 1.117 V for positive saturation). Figure 19 tracks this transient response, showing that the sensor settles to within 5 mV of its new baseline in 40.1 ms after the negative event, and 48.7 ms after the positive event.

Cumulative Baseline Shift: Figure 20 illustrates the permanent offset introduced by each event. The sensor's initial resting baseline of -4.8 mV shifted up slightly to -1.6 mV following the negative saturation. However, the subsequent positive saturation caused a massive upward shift, leaving the final resting baseline at 11.3 mV.

Gradient Stability: Despite a total zero-current offset shift of 16.05 mV, the sensor's linear response to current is essentially unaffected. Table 14 confirms that the gradient drops by a negligible -0.5

Table 14: Fresh sensor: linear fit of the current sweep before and after the saturation events in Table 13.

Sweep	Gradient (mV/A)	Offset at 0 A (mV)	R^2
Before saturation	1.289	−4.55	0.991
After saturation	1.283	+11.50	0.994
Change	−0.5%	+16.05	

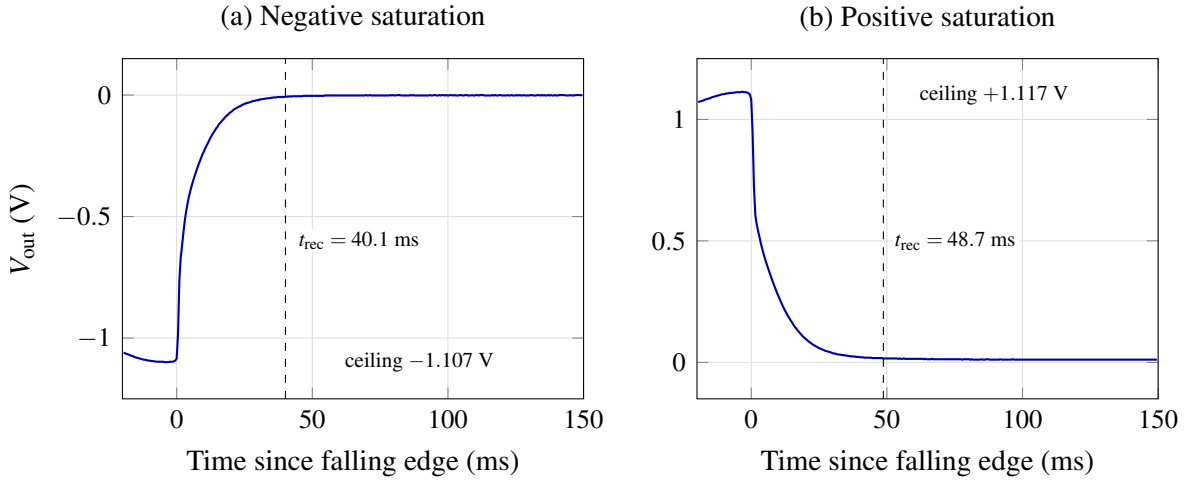


Figure 19: Saturation recovery on the fresh sensor after the magnet is removed, for (a) negative and (b) positive saturation. Time zero is the falling edge; the dashed line marks the recovery time (output stays within 5 mV of its final baseline).

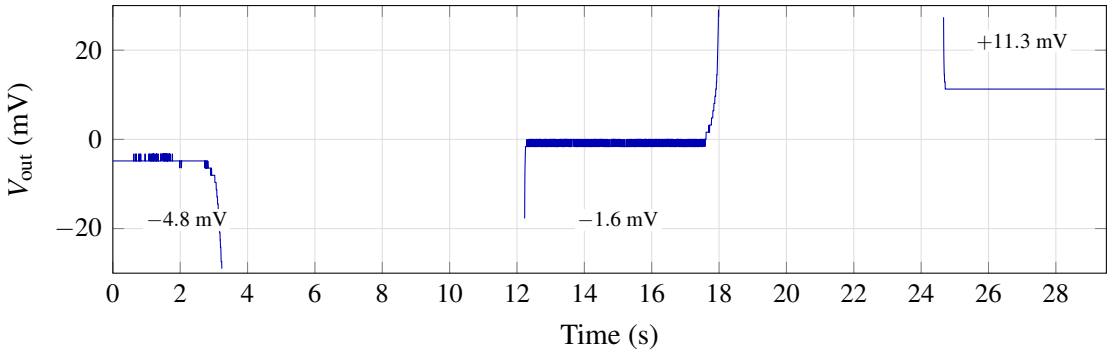


Figure 20: Output baseline before, between and after the two saturation events (view clipped to ± 30 mV so the baseline is visible). The baseline moves from -4.8 mV to -1.6 mV after the negative saturation, then to $+11.3$ mV after the positive saturation.

12 Discussion

The experimental results confirm several of the CT100's key design properties and surface a small number of practical issues to resolve before the hardware can be used as a precision, repeatable test rig.

Bridge topology and saturation mechanism. The $V_{CC}/3$ saturation ceiling is not merely an empirical observation; Section 4.3 shows it follows directly from the CT100's specified 20–40 kΩ per-element resistance range via Equation (**??**), and the measured value (1.12 V at $V_{CC} = 3.30$ V) matches this derivation closely. This is a useful confirmation: the sensor's saturation ceiling can be predicted from V_{CC} alone for any future supply condition, without needing to empirically re-characterise it each time.

Field-model discrepancy. The Ansys and analytical models agree closely with each other but under-predict the measured sensor output by roughly a factor of two, and the fitted sensitivity (Section 8.2) sits about 18% below the datasheet figure. Because the sensitivity result is a slope and therefore immune to a fixed additive offset, both discrepancies point toward the same explanation: ambient background field pickup in the absence of shielding, rather than an error in the trace model itself. The preliminary shielded results (Section **??**) moving closer to the simulated prediction support this explanation directly.

Saturation recovery and ADCS implications. The measured 82.3 ms recovery time (Section 8.1) is directly relevant to how closely a magnetorquer firing and a magnetometer reading can be scheduled without corrupting the ADCS control loop. **[TODO: If a specific safe guardband G has been calculated against a target OBC sampling frequency, insert that comparison and result here.]**

Hysteresis. The observation that saturation leaves a permanent, linear-but-offset baseline (Section 6.3) is a practically important finding for any CubeSat deployment: it implies the sensor's zero-point cannot be assumed stable across the mission if it experiences even one strong saturating event, and would need either periodic re-calibration or a design that avoids saturating the sensor in the first place. **[TODO: This finding needs the quantification and reconciliation noted in Section 6.3 before it can be stated this firmly.]**

Radiation tolerance. The post-irradiation sensitivity sweep (Section 9.4) measured 93.6% (DMM) and 99.8% (MCU) of the unirradiated sensitivity, with no consistent direction of change

between two independent instruments, supporting the literature consensus that MTJ-based TMR sensors tolerate ionising radiation substantially better than conventional CMOS devices. A single irradiated unit, tested once, should still be presented as preliminary evidence; a second, independent irradiated replicate would substantially strengthen this conclusion (Section 13).

[TODO: write about rdaition 100krad or whatever talk about litertry what they did what i did, kinda verfiyies what they did we did a smaller dose of a different radation and and did a COTS device]

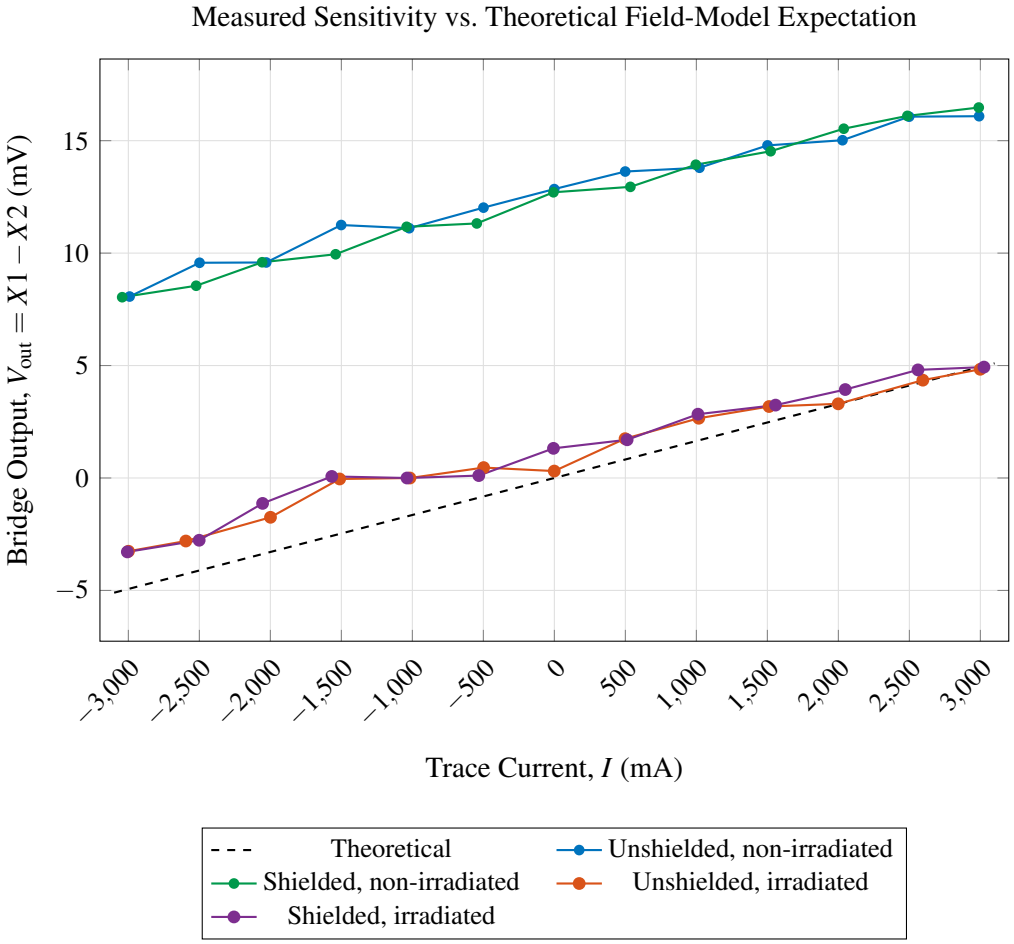


Figure 21: Measured bridge output vs. trace current across all four test conditions, against the theoretical field-model expectation (CT100 datasheet sensitivity of 4.5 mV V^{-1} mT^{-1}). All four measured conditions fall below the theoretical line, consistent with the ambient-field-driven discrepancy discussed in this section; the shielded, non-irradiated condition sits closest to it. Note each dataset carries its own session-specific baseline offset here; see Figure **??** for a baseline-corrected, slope-only comparison.

12.1 Methodological Limitations and Future Experimental Refinements

[TODO: add radaition section values in and the get section names right] It should be noted that the measured transient recovery time ($t_{rec} \approx 73.5$ ms–85.0 ms) for the non- irradated and xxxx trec for radiation in Section 6.2 and in Section 6.2 respectively is not a repeatable or automatable stimulus represents an upper bound dictated by the experimental setup rather than the intrinsic response limit of the TMR junction. Because the external magnet was removed

manually, the observed decay trajectory is dominated by human operator speed and the resulting rate of field decay (dB/dt). In reality, the fundamental quantum spin-tunneling dynamics of Magnetic Tunnel Junctions operate on microsecond to nanosecond timescales, suggesting that the CT100's intrinsic response is substantially faster than observed.

To isolate the sensor's physical relaxation kinetics from operator motion artifacts, future characterization requires an automated, high-bandwidth stimulus system. Recommended alternative methodologies include:

- **Pulsed Trace Excitation:** Driving high-speed current pulses through the integrated PCB trace using a low-inductance MOSFET switching circuit to generate microsecond-scale magnetic step functions.
- **Switched Helmholtz Coil:** Utilizing a low-inductance air-core Helmholtz coil driven by a pulse generator to deliver controlled, highly repeatable transient magnetic steps without mechanical motion.

13 Conclusions and Future Work

Key findings.

- The CT100's $V_{CC}/3$ saturation ceiling is derived directly from its specified per-element resistance range (20–40 kΩ) and confirmed experimentally (1.12 V measured at $V_{CC} = 3.30\,\text{V}$).
- Saturation leaves the sensor at a new, offset-but-linear baseline rather than the original zero-field value, a hysteresis effect with direct implications for in-mission recalibration. **[TODO: Pending quantification – see Section 6.3.]**
- Measured sensitivity is $3.7\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}$, approximately 18% below the CT100's $4.5\,\text{mV}\,\text{V}^{-1}\,\text{mT}^{-1}$ datasheet figure, most plausibly attributable to ambient-field pickup in the absence of shielding rather than a sensor shortfall.
- Saturation recovery on the integrated board takes 82.3 ms ($\approx 12.15\,\text{Hz}$), directly relevant to ADCS guardband design around magnetorquer firings.
- Radiation testing found no statistically significant change in sensitivity pre- vs. post-irradiation (93.6–99.8% of the unirradiated value), a preliminary but potentially novel data point for this specific exposure regime.
- An initial shielded enclosure showed a qualitative improvement toward simulated field predictions, supporting ambient pickup as the cause of the field-model discrepancy above.

Future work.

- **Improve the saturation stimulus.** The hand-held magnet is acknowledged as a limiting factor throughout this report. Given that radiation has now been shown to have no significant effect on sensitivity, remaining time is better spent developing a more repeatable, automated saturation source (e.g. a driven electromagnet or Maxwell coil) and re-running the full characterisation on the non-irradiated integrated board, rather than further radiation testing.

- Complete and quantify the shielded and radiation-plus-shielded sweeps (Sections **??** and **??**).
- Quantify the hysteresis offset magnitude and confirm its stability over repeated saturation cycles.
- Repeat the saturation-recovery measurement multiple times to report a mean $\pm$ standard deviation rather than a single trial, and add a hysteresis (up/down current sweep) linearity measurement.
- Obtain a second, independent irradiated replicate, ideally on an unreworked board, to strengthen the single-unit radiation-tolerance conclusion and control for the resolder-cycle limitation noted in Section 12.
- Implement the non-zero reference-bias design change proposed in Section 7.6 on a future PCB revision.

Overall conclusion. Based on the evidence gathered to date, the CT100 TMR sensor appears viable for CubeSat ADCS/EPS applications: its bridge behaviour is well-characterised and predictable, its sensitivity is within a reasonable margin of its datasheet specification once ambient-field effects are accounted for, and it shows no measurable radiation-tolerance penalty at the dose tested. This should be read as an encouraging preliminary result rather than a qualification conclusion; the future work above, particularly a repeatable saturation source and a second irradiated replicate, would be needed to state this with confidence.

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